

CS 609: Speech Processing

Course Code	CS 609
Course Name	Speech Processing
L-T-P-C	3-0-2-4
Prerequisites	Signals and Systems (IC260), or Digital Signal Processing (EE305) or COT
Intended for	BTech./MTech./M.S./Ph.D.
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introduction** [1 Lecture]
- **Review of digital signal processing:** Discrete-time signals and systems, transform representation of signals and systems, fundamentals of digital filters, sampling. [3 Lectures]
- **Fundamentals of human speech production, acoustic theory of speech production:** Speech production, short-time Fourier transform, acoustic phonetics. [3 Lectures]
- **Hearing and speech perception, auditory models:** Anatomy and functions of the ear, the perception of sound, auditory models, lossless tube models. [3 Lectures]
- **Time-domain methods:** Short-time analysis (energy, magnitude, zero-crossing rate, autocorrelation). [5 Lectures]
- **Frequency-domain methods:** Discrete-time Fourier analysis, short-time Fourier analysis, spectrograms, Overlap-add method of synthesis, filter-bank summation method of synthesis. [7 Lectures]
- **Cepstrum and homomorphic speech processing:** Homomorphic analysis, computing the short-time cepstrum and the complex cepstrum, cepstrum analysis of all-pole models, cepstrum distance measures. [5 Lectures]
- **Linear predictive analysis of speech:** Basic ideas, gain computation, frequency-domain interpretation, solving LPC equations, the prediction error signal, representations of LP parameters. [6 Lectures]
- **Algorithms for estimating speech parameters:** Median smoothing, speech-background discrimination, pitch period estimation, formant estimation. [5 Lectures]
- **Digital coding of speech signals:** Sampling speech signals, statistical models for speech signals, quantization (instantaneous, adaptive), quantising speech model parameters, delta modulation, DPCM, AD-PCM. [3 Lectures]
- **Applications: speech recognition, speech enhancement, speaker recognition:** Hidden Markov models for speech recognition, statistical methods for speech enhancement, factor analysis for speaker recognition. [4 Lectures]

Text Books

1. L.R. Rabiner, R. W. Schafer, **Theory and applications of digital speech processing**, Prentice Hall
2. L.R. Rabiner, R. W. Schafer, **Digital Processing of Speech Signals**, Pearson
3. Douglas O'Shaughnessy, **Speech Communications: Human and Machine**, Wiley India
4. Ben Gold and Nelson Morgan, **Speech and Audio Signal Processing: Processing and Perception of Speech and Music**, Wiley

References

1. J. R. Deller, J. H. L. Hansen, J. G Proakis, **Discrete-time processing of speech signals**, Wiley.
2. A. V. Oppenheim, R. W. Schafer, R. Buck, **Discrete-time signal processing**, Pearson